PROJECT REPORT TITLE:

CSE DEPARTMENT PORTAL

SUBJECT:FST

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1. Introduction

The CSE Department Portal is a comprehensive full-stack web application designed to digitize and streamline the academic and administrative workflow within a Computer Science and Engineering department.

Traditionally, lab scheduling, assignment tracking, and communication between faculty and students are handled manually or through fragmented systems. This leads to inefficiencies, scheduling conflicts, and lack of transparency.

This project addresses these issues by providing a centralized digital platform where all stakeholders—including Students, Faculty, Class Representatives (CRs), and Administrators—can interact efficiently.

The portal integrates multiple functionalities such as:

- Lab booking and scheduling
- Task and assignment management
- Announcement broadcasting
- Role-based dashboards

Additionally, the system is built using both MERN (React) and MEAN (Angular) stacks, demonstrating multi-framework compatibility with a shared backend.

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2. Objective

The main objectives of this project are:

- To develop a centralized academic management system
- To eliminate manual lab booking processes
- To prevent scheduling conflicts using automated logic
- To enable efficient communication through announcements
- To provide role-based dashboards for better usability
- To demonstrate the use of multiple frontend frameworks with a single backend
- To improve student engagement and tracking of academic progress

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3. Technologies Used

Frontend Technologies

◈ React.js (MERN)

- Component-based UI
- Fast rendering using Virtual DOM
- Uses Hooks like:
    - useState
    - useEffect
    - useRef
    - useMemo

◈ Angular (MEAN)

- Structured framework
- Uses Signals for state management
- Supports:
    - Dependency Injection
    - Modular architecture
    - Two-way data binding

Backend Technologies

◈ Node.js

- Event-driven architecture
- Non-blocking I/O operations

◈ Express.js

- Lightweight framework for building APIs
- Middleware support

Database

◈ MongoDB

- NoSQL database
- Flexible schema
- High scalability

Security & Tools

- JWT Authentication
- Nodemailer (Email system)
- Helmet (Security headers)
- Rate Limiting (API protection)

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4. System Architecture

The system follows a three-tier architecture:

1. Presentation Layer
   - React frontend
   - Angular frontend
2. Application Layer
   - REST APIs using Express
3. Data Layer
   - MongoDB database

Both frontends communicate with the same backend, ensuring consistency and scalability.

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5. Modules of the System

◈ Authentication Module

- User registration and login
- Role assignment (Admin, CR, Student, Faculty)
- Secure authentication using JWT

◈ Lab Management Module

- CR can request lab bookings
- Select lab, date, and time slots
- Conflict detection prevents overlapping bookings
- Admin approval system

◈ Task/Assignment Module

- Faculty creates assignments
- Students submit tasks
- Status tracking (Pending/Completed)

◈ Announcement Module

- Admin posts announcements
- Optional email notifications
- Visible on dashboard

◈ Academic Module

- Student profile
- Course tracking
- Academic details

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6. Components Used

React Components

- Navbar
- Dashboard
- Task Pages
- Lab Request Page

Angular Components

- Dashboard Component
- Task Component
- Lab Component

Reusable Components

- Buttons
- Cards
- Forms
- Modals

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7. Functional Components

React Features

- State Management using Hooks
- Lifecycle handling
- Performance optimization

Angular Features

- Signals for reactive state
- HttpClient for API calls
- Routing and navigation

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8. Database Design

User Collection

Stores user details and roles.

Task Collection

Stores assignments and status.

Booking Collection

Stores lab booking data.

Announcement Collection

Stores messages and updates.

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9. Working of the System

User logs into system

1. Dashboard loads based on role
2. CR requests lab booking
3. System checks for conflicts
4. Admin approves/rejects request
5. Tasks and announcements are managed
6. Data updates dynamically

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10. Features

- Role-based access
- Lab booking system
- Task management
- Announcement system
- Conflict detection
- Dual frontend architecture

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11. Advantages

- Reduces manual workload
- Improves efficiency
- Real-time updates
- Scalable system
- Better communication

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12. Limitations

- Internet dependency
- Email configuration required
- Increased complexity due to dual frontend

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13. Future Enhancements

- AI-based scheduling
- Mobile app version
- QR attendance system
- Real-time notifications

14. Conclusion

The CSE Department Portal successfully provides a unified solution for academic and administrative management.

By combining MERN and MEAN stacks, the system demonstrates flexibility, scalability, and modern development practices.

It enhances efficiency, reduces manual errors, and improves communication within the department, making it a valuable tool for real-world implementation.